INCH-POUND
MIL-R-55342E
AMENDMENT 5
25 July 1994
SUPERSEDING
AMENDMENT 4
9 MAY 1994

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, CHIP,
ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-55342E, dated 26 January 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE II (excluding footnotes), delete and substitute:

"TABLE II. Termination material.

Type	Material	Termination area	Code letters
Solderable 1/ Bondable Weldable	Gold	Wrap around 2/ One surface Bonding pads	G W 3/ P
Solderable 1/	Base metallization barrier metal, solder coated	Wrap around 2/	B
	Pretinned	Wrap around 2/ One surface	R 4/ S 3/
Bondable weldable	Platinum/gold	Wrap around One surface	U T 3/
Bondable	Palladium/silver or Platinum/silver	Wrap around	C
	Palladium/silver or Platinum/silver	One surface	D 3/

"

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2.1.2, add the following:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 - Standard Specification for Electrodeposited Coating of Tin.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)"

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PAGE 10

3.3.2.1, add the following sentence:

"In addition, the manufacturer shall demonstrate resistance temperature characteristic (RTC) control in the process."

PAGE 11

After 3.5.2.1, add new paragraph:

"3.5.2.2 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.7.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

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3.20.3, delete and substitute:

"3.20.3 Supplying to higher failure rate levels. A manufacturer may supply to all higher failure rate levels than to which he is qualified. Parts qualified to lower failure levels, with procuring agency approval, are substitutable as long as the label reflects the original failure rate level ordered (see table VIII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2)."

TABLE VIII, delete and substitute:

"TABLE VIII. Failure rate substitution.

Failure rate	Failure rate substitute
S (.001)	.....
R (.01)	S
P (0.1)	S, R
M (1.0)	S, R, P

"

3.20.4, delete and substitute:

"3.20.4 Supplying to lower resistance temperature characteristics and/or lower resistance tolerances. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels are substitutable, with procuring agency approval, for higher resistance temperature or higher tolerance levels as long as the label reflects the original resistance temperature characteristic and/or resistance tolerance ordered (see tables IX and X). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2)."

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TABLE IX, delete and substitute:

"TABLE IX. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitute
0.1.....	.....
0.5.....	0.1
1.0.....	0.1, 0.5
2.0.....	0.1, 0.5, 1.0
5.0.....	0.1, 0.5, 1.0, 2.0
10.0.....	0.1, 0.5, 1.0, 2.0, 5.0

"

* TABLE X, delete and substitute:

"TABLE X. Characteristic substitution.

Characteristic	Characteristic substitute
Y	.....
E	Y
H	Y, E
K	.....
M	K

"

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4.4.4.2, last sentence, delete.

4.5h, last sentence, delete "detail specification" and substitute "PIN".

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4.6.1, add the following new sentence:

"Group B inspection for preparation of delivery is not required when the qualifying activity has allowed group B testing to be performed annually (see table XIV)."

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Add the following new paragraphs:

"4.6.1.2.4 Subgroup 4 (solderability).

"4.6.1.2.4.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

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"4.6.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use the following option to rework the lot:

Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.11. An additional 13 samples from each production lot shall be tested. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification. Individual production lots that pass the solderability test are available for shipment.

"4.6.1.2.4.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

TABLE XII, add the following new subgroup:

"

Subgroup 4			
Solderability	3.17	4.7.11	See 4.6.1.2.4

"

TABLE XIII, delete and substitute:

" TABLE XIII. Sampling plans for PPM categories.

Lot size	Sample size PPM-2	Sample size PPM-3
1 - 13	100%	100%
14 - 125	100%	13
126 - 150	125	13
151 - 280	125	20
281 - 500	125	29
501 - 1200	125	34
1201 - 3200	125	42
3201 - 10,000	192	50
10,001 - 35,000	294	60
35,001 - 150,000	294	74
150,001 - 500,000	345	90
500,001 and above	435	102

"

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TABLE XIV, delete and substitute (change from amendment 1):

" TABLE XIV. Group B inspection.

Inspection	Requirements paragraph	Method paragraph	Number of samples
Subgroup 1 1/			
Resistance temperature characteristic	3.15	4.7.9	13
Short time overload	3.11	4.7.5	
Subgroup 2 2/			
Adhesion	3.18	4.7.12	13

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity can be performed on a annual basis. If the design, material, construction or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failures, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every three years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed or, if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute."

4.6.1.3.1.3, delete in its entirety.

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TABLE XV, delete and substitute:

"TABLE XV. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Monthly				
Subgroup 1				
Life	3.16	4.7.10	See 4.6.2.1.1.1	See 4.6.2.1.1.1
Subgroup 2 1/				
Thermal shock	3.9	4.7.3	30	1
Low temperature operation	3.10	4.7.4		
Quarterly				
Subgroup 1				
Resistance to bonding exposure	3.13	4.7.7	30	1
Moisture resistance	3.14	4.7.8		
Subgroup 2 1/				
High temperature exposure	3.12	4.7.6	30	

1/ If the manufacturer can demonstrate that these tests have been performed five consecutive times with zero failures, the frequency of these tests, with the approval of the qualifying activity can be performed on a annual basis. If the design, material, construction or processing of the part is changed or, if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency."

PAGE 33

4.7.1.3b., delete and substitute:

"b. Termination material "W" and "P" (weldable). Chip resistors with weldable terminations shall be mechanically attached to the test substrate and electrically connected by wire bonding flying lead interconnections between the chip termination and the test substrate metallized land areas. The interconnecting lead shall be .001 inch (0.03 mm) in diameter gold wire. The chip shall be mechanically mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test."

PAGE 34

4.7.4, fourth sentence, delete "for a period of approximately 24 hours" and substitute "until stabilized".

4.7.6, first sentence, delete and substitute:

"Resistors shall be mounted as specified in 4.7.1.3 and the dc resistance shall be measured as specified in 4.7.2."

4.7.7, third sentence, delete "for 4 to 12 hours".

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4.7.8c, delete as substituted in Amendment 3.

4.7.9, add the following new subparagraph:

"e. At the option of the manufacturer, the reverse sequence of table XVII may be as follows:

1 - Room temperature.
2 - Series of hot temperatures.
3 - Room temperature.
4 - Series of cold temperatures for the subgroup 1 tests."

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After 6.7, add the following new paragraph:

"6.7.1 Tin plated finishes. Tin plating is prohibited (see 3.5.2.2) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

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* TABLE XX, delete and substitute:

"TABLE XX. Extension of qualification.

Style	Will qualify style	
RM0502	RM0502	
RM0505	RM0505	
RM0705	RM0705, RM0505	
RM1005	RM1005, RM0705, RM0505	
RM1010	RM1010, RM1005, RM0705, RM0505	
RM1505	RM1505, RM1005, RM0705, RM0505	
RM2208	RM2208	
RM1206	RM1206, RM1505, RM1005, RM0705, RM0505	
RM2010	RM2208, RM2010	
RM2512	RM2010, RM2208, RM2512	
Characteristic	Will qualify characteristic	May qualify 1/ characteristic with minor additional testing
Y	Y, E, H	K, M
E	E, H	K, M
H	H	K, M
K	K, M	----
M	M	----
Termination	Will qualify termination	May qualify 1/ termination with minor additional testing
G	G, W, P	B, R, S
W	W, P	S
P	P, W	S
B	B, R, S	G, W, P 2/
R	R, S	-------
S	S	-------
U	U, T	C, D
T	T	D
C	C, D	U, T
D	D	T
Failure rate level	Will qualify failure rate level	
S	S, R, P, M	
R	R, P, M	
P	P, M	
M	M	
Resistance tolerance	Will qualify resistance tolerance	
0.1	0.1, 1.0, 2.0, 5.0, 10.0	
1.0	1.0, 2.0, 5.0, 10.0	
2.0	2.0, 5.0, 10.0	
5.0	5.0, 10.0	
10.0	10.0	

1/ To be determined by qualifying activity.
2/ Must contain barrier layer to qualify."

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CONCLUDING MATERIAL

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Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME
Navy - AS, CG, MC, OS
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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